An Introduction to Pneumatics Systems

Basics of Pneumatics and Pneumatics Actuator Systems

Learning Objectives

- What will you know/be able to do by the end?
  - Explain basic pneumatics principles
  - Identify major pneumatic system parts
  - Able to create simple pneumatics circuits

Pneumatics

- **Pneumatics** (from Greek πνεῦμα *pneuma* 'wind, breath') is the use of gas or pressurized air in mechanical systems.

- The idea of using compressed air to carry out work goes back a long way to the Roman era.
- Heron of Alexandia, also known as Hero of Alexandria, a Greek mathematician and engineer created automatically opening temple doors mechanism.



The book About automata by Hero of Alexandria (PC: Wikipedia)

Pneumatics = Controlled flow of pressurized air

Pneumatics is the technology process starting with the **generation** of compressed air, followed by its **preparation**, and finally its **distribution**

Pressure, Force, and Flow in Pneumatics

- Pneumatic systems operate based on three central ideas; **pressure**, **flow**, and **force.**

Pressure:

- It represents the **stored energy within the system**, waiting to be released to perform work.
- The higher the pressure, the more energy is available to drive pneumatic devices.

Force:

- The force applied determines the **mechanical output of the actuator,** with force being the product of pressure and surface area, resulting in linear or rotary motion.
- Higher the force, **stronger actuator**

$F = P \; X \; A$ **(Force = Pressure × Area)**

Flow:

- Flow rate determines how fast air moves through the system, affecting **the speed** at which pneumatic devices, such as actuators, perform tasks.
- A higher volume of air flow results **in faster actuators.**

Flow (*Q*)=Volume/Time

Pressure, Force, and Flow in Pneumatics

- Pneumatic systems operate based on three central ideas; **pressure**, **flow**, and **force.**

Pressure: Pressure is the force exerted per unit area, typically measured in pounds per square inch (psi) or bars.

Role:

- It represents the **stored energy within the system**, waiting to be released to perform work.
- The higher the pressure, the more energy is available to drive pneumatic devices.

Force: Force is the push or pull created by the pneumatic actuator, resulting from pressure applied to a specific surface area.

Role:

- The force applied **determines the mechanical output** of the actuator, with force being the product of pressure and surface area, resulting in linear or rotary motion.

Flow: Flow refers to the volume of air passing through a specific area within a given time frame.

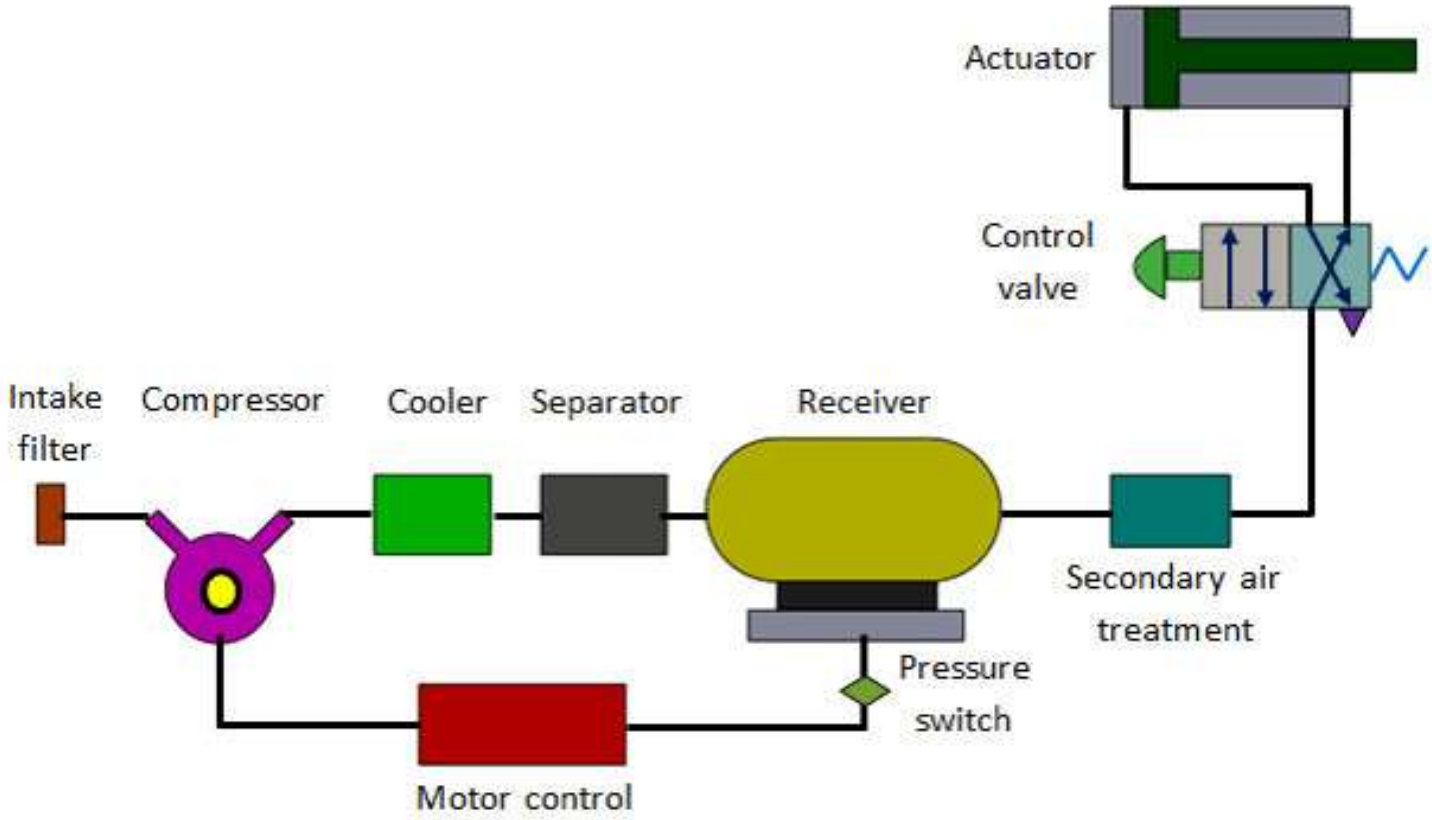
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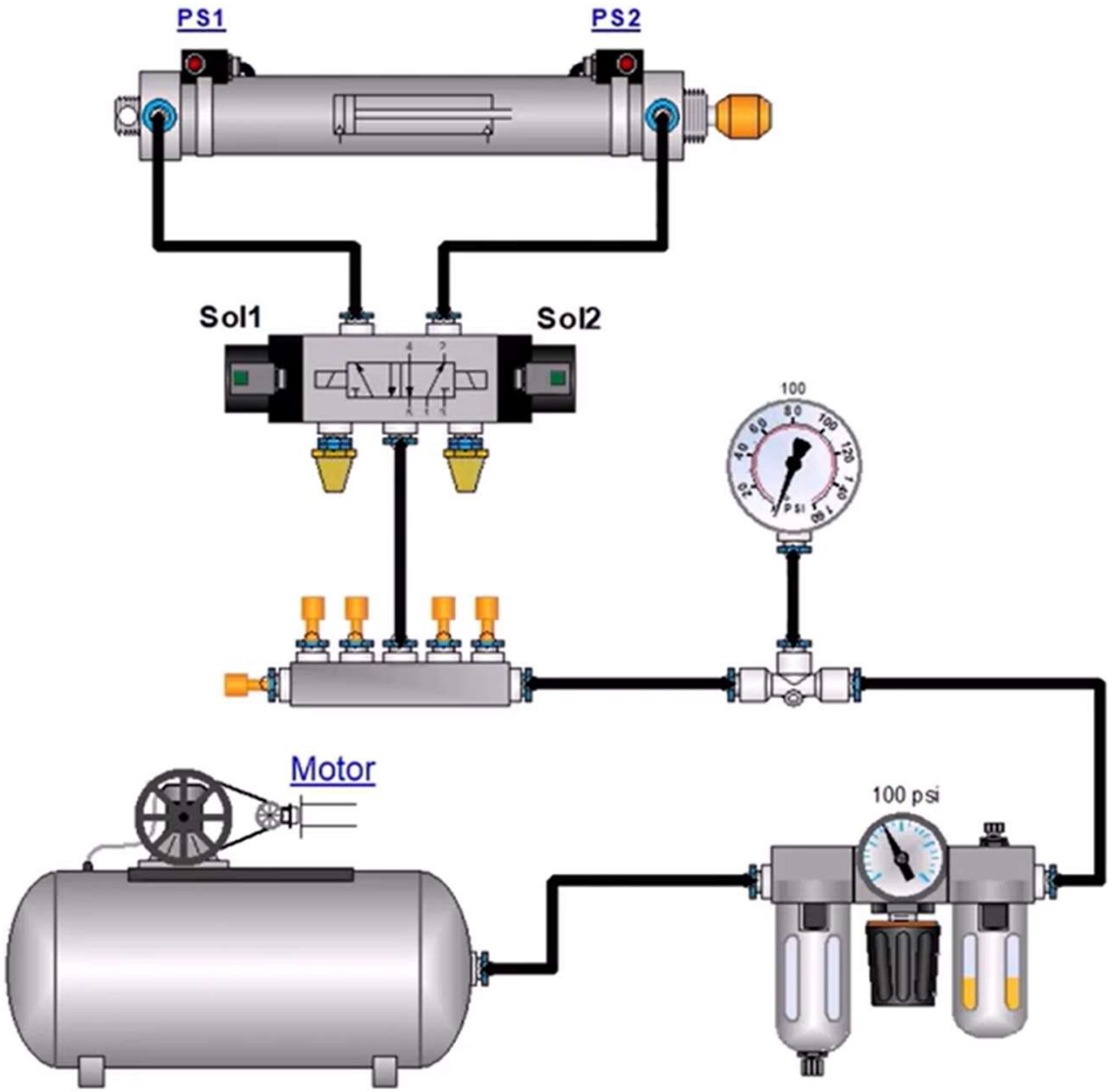
Pneumatic System Overview

Main subsystems:

- **Air generation and conditioning** – prepares clean, pressurized air.
  - Compressor, Air receiver tank etc.
- **Control valves** – regulate direction, pressure, and flow.
- **Actuators** – convert air pressure into motion (linear or rotary).



Pneumatic** systems are like **hydraulic systems, but** they use **compressed air** instead of **a fluid to transmit power



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Air Generation and Storage System

Air compressor:

- The compressor is the **primary source** of energy in a pneumatic system.
- The air compressor **draws air from the atmosphere**, **pressurizes it** and stores the compressed air in a tank for release to the transmission system.
- It converts mechanical energy (from an electric motor or engine) into potential energy stored in compressed air.



Air Compressor with Horizontal Tank



Air Compressor with Vertical Tank

Air tank (air receiver tank) - is a pressure vessel that stores compressed air and functions as a buffer between the air compressor and downstream equipment .

A tank holds compressed air and distributes through pipes to maintain a steady supply.



Air Preparation/ conditioning System

Air Preparation (FRL Unit): The untreated air from a compressor is typically hot, dirty, and wet, the FRL unit cleans, conditions, and lubricates it to ensure efficient performance and prolong the life of pneumatic tools and components.



- Cleans, regulates, and conditions air for reliable and long component life.
- **Filter (F):** The filter is the first line of defense against contaminants. It separates and removes harmful debris that can accumulate in compressed air lines. It removes moisture, dust, oil particles.
- **Regulator (R):** The regulator controls and maintains a constant air pressure output for the downstream equipment.
- **Lubricator (L):** The lubricator adds a controlled amount of oil mist into the airflow to reduce friction in moving components.
  - Helps to reduce wear, prevent corrosion, and maximize the operational life of tools and equipment.

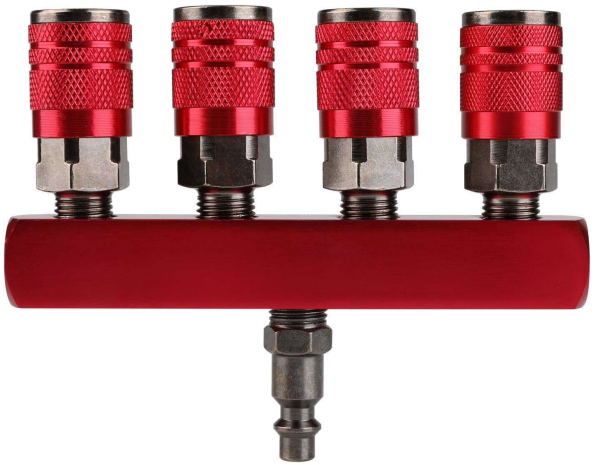
Air Distribution System

- Transports compressed air from the supply to control and actuator units.
  - Pipes, hoses, fittings
  - Manifolds, connectors





Air Manifold

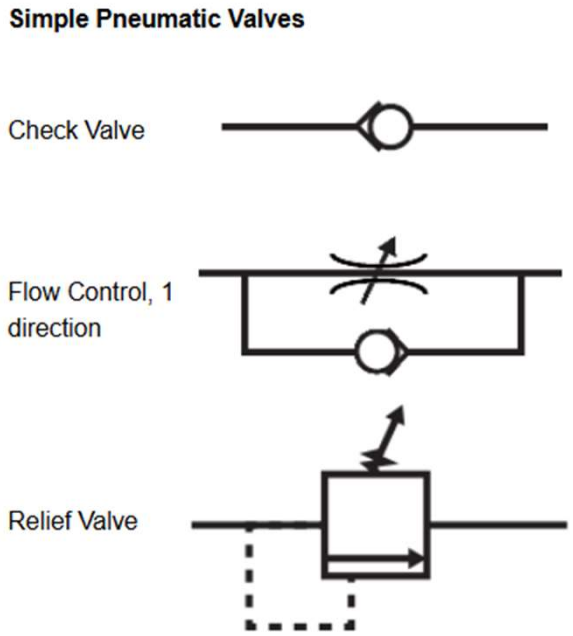


4-Way Straight Air Manifold 5 Ports

Control Valves

Controls **direction, flow rate, and pressure** of air to actuators.

- **Check Valves:** A check valve is a self-operating, one-way valve that allows fluid (liquid or gas) to flow in a single direction while preventing backflow
- **Flow Control Valve:** adjusts actuator speed by controlling exhaust airflow.
- **Pressure Relief Valve:** protects the system from over-pressure.
- **Directional Control Valve (DCV):** directs air to extend or retract cylinder (e.g., 5/2 valve, check valve).

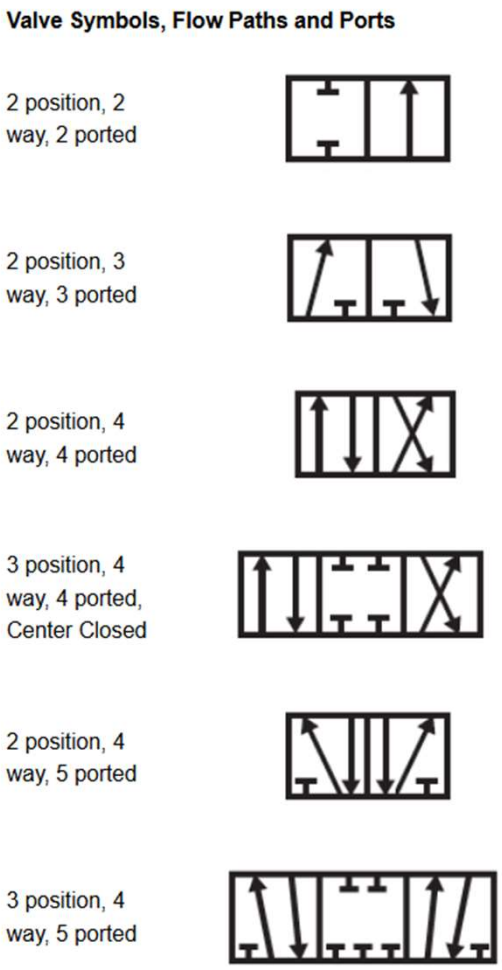


Direction Control Valves

- DCVs are categorized by the number of **ports and positions** they have, as well as their actuation method.

- Valves are also referred to by the number of "ways" that air can enter or exit the valve.

  - **2/2 – way Valve**
  - **3/2 – way Valve**
  - **4/2 –way Valve**
  - **4/3 – way Valve**
  - **….**

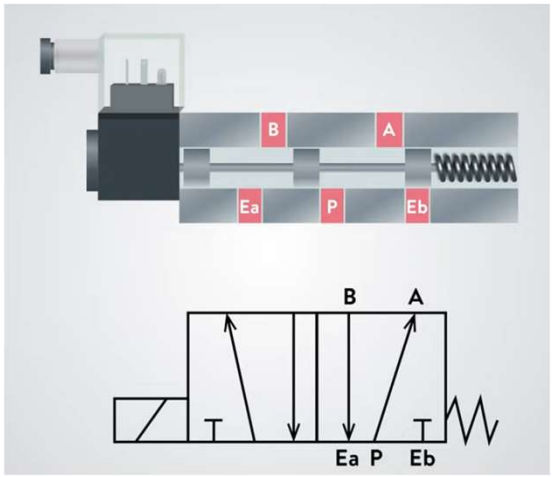


Examples of DCV types and actuations

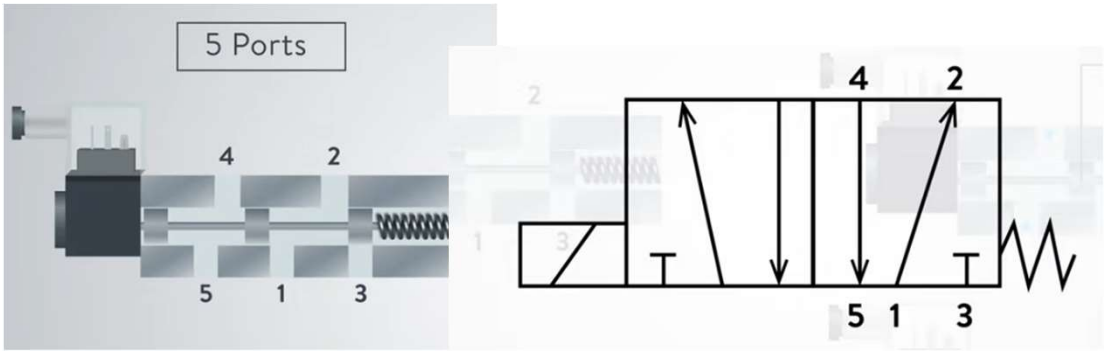
Ports vs "Ways"

A **port** is a physical connection point on the valve body, while a **way** is a functional pathway that air can travel through the valve.

- This valve has **five ports** (P, A, B, Ea, Eb) , but it is sometimes referred to as a 4-way valve in **older (hydraulic) terminology** because two **exhaust ports (Ea and Eb) share the same function**, they both vent to atmosphere and are counted as one functional pathway.



- **In modern pneumatics**, each port is counted individually, so it is more accurately described as a 5-port, 2-position valve (commonly written as 5/2 valve). The two exhausts (R and S) allow independent venting from ports A and B, improving cylinder control.

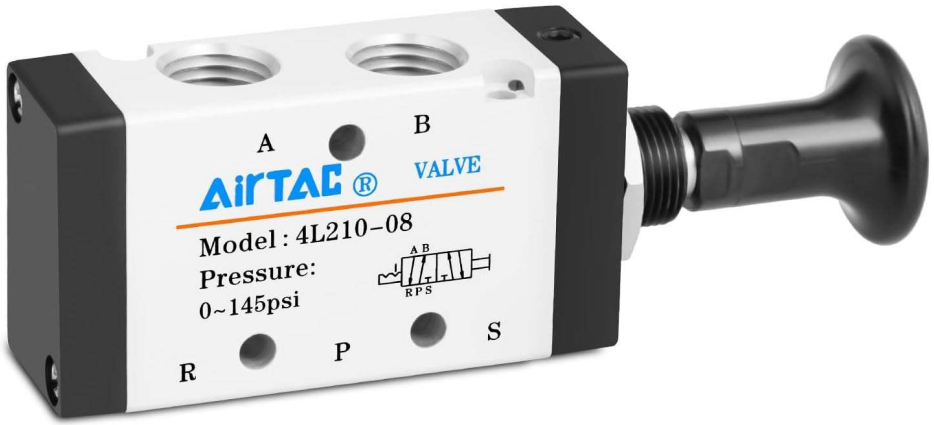


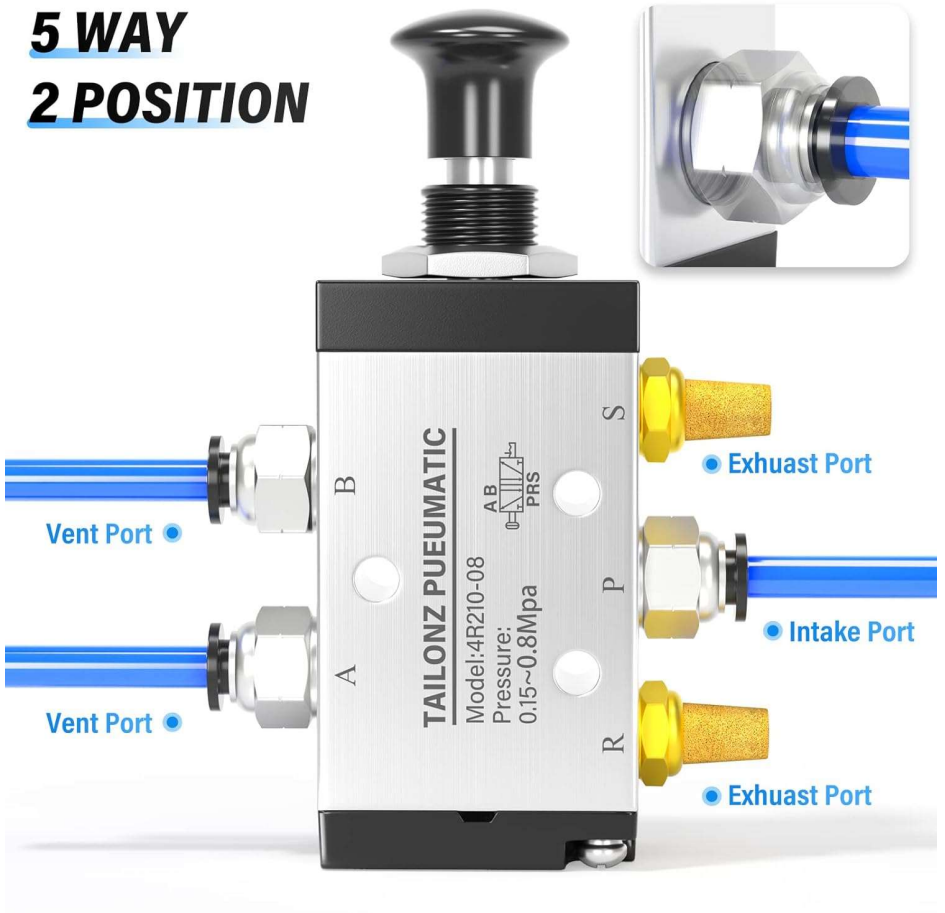
What type of Valve is this?



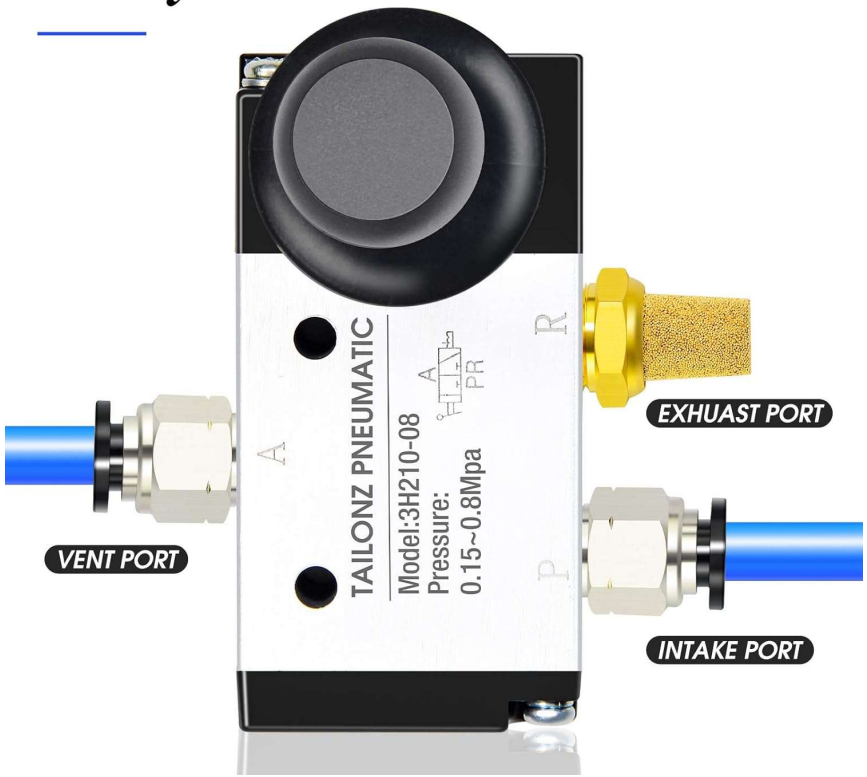
Check it

5/2 Valve


A
B
AirTAC ®
VALVE
Model : 4L210-08
Pressure:
0~145psi
R
P
S


5 WAY
2 POSITION
Vent Port
Vent Port
Exhuast Port
Intake Port
Exhuast Port
TAILONZ PUEUMATIC
Model:4R210-08
Pressure:
0.15~0.8Mpa
A
B
P
R
S

3 Way 2 Postion

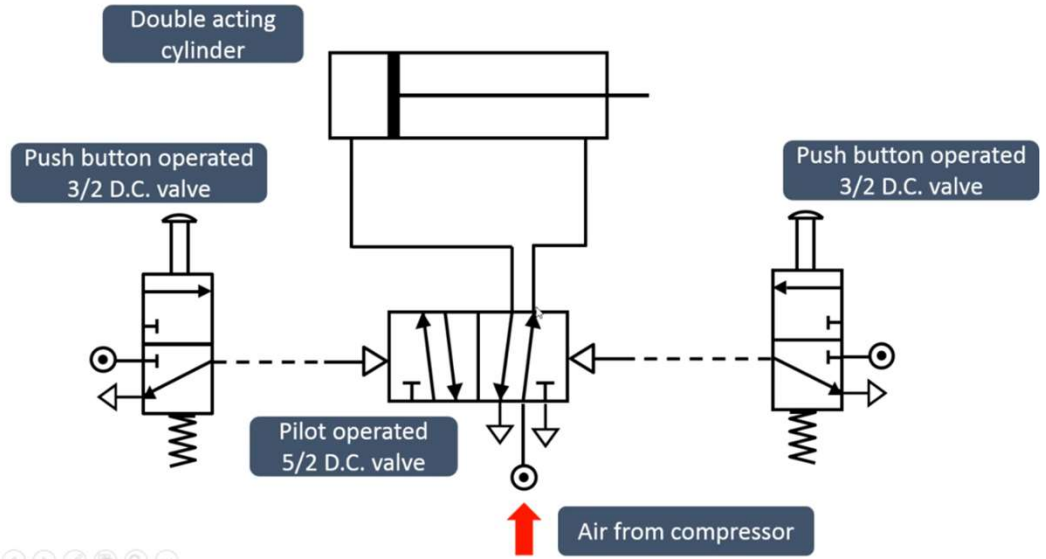


Actuation Methods of DCV

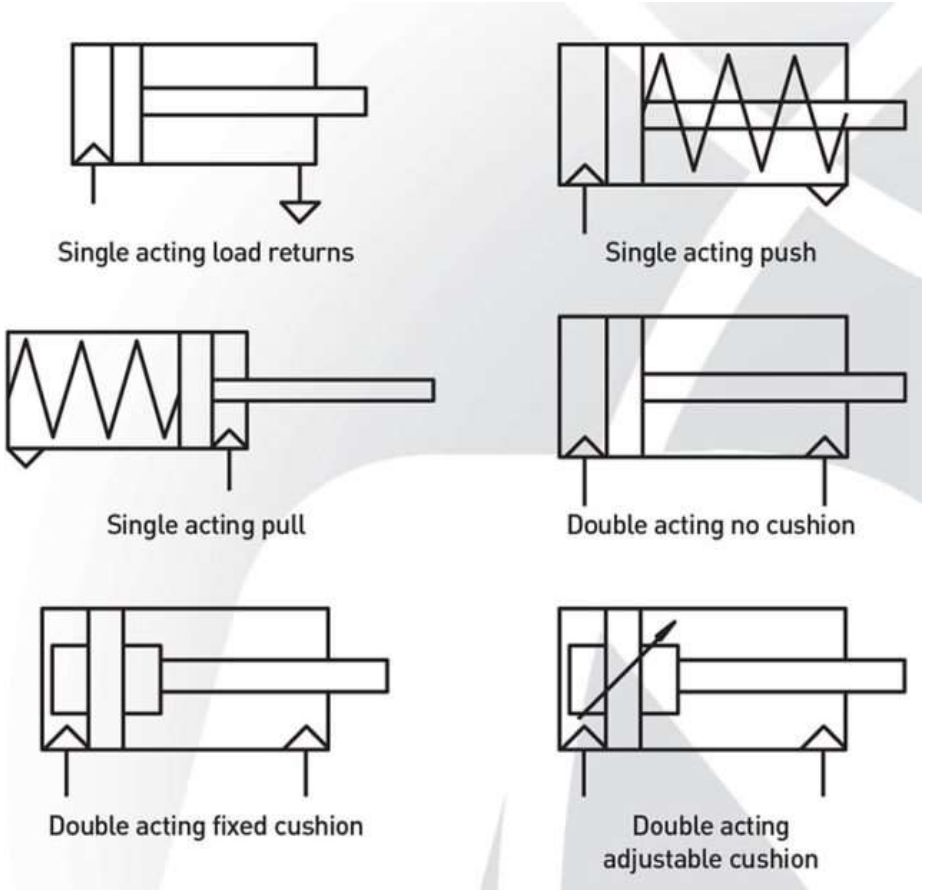
Types of Actuation

Push Button Lever	**Manual Actuation**
Solenoid	**Electrical Actuation**
Spring Roller lever Foot pedal	**Mechanical Actuation**
Pneumatic	**Pneumatic & Hydraulic Actuation**
Hydraulic	

Pilot actuation method uses a smaller, lower-power valve (the pilot) to control the shifting of a larger, higher-power main valve.

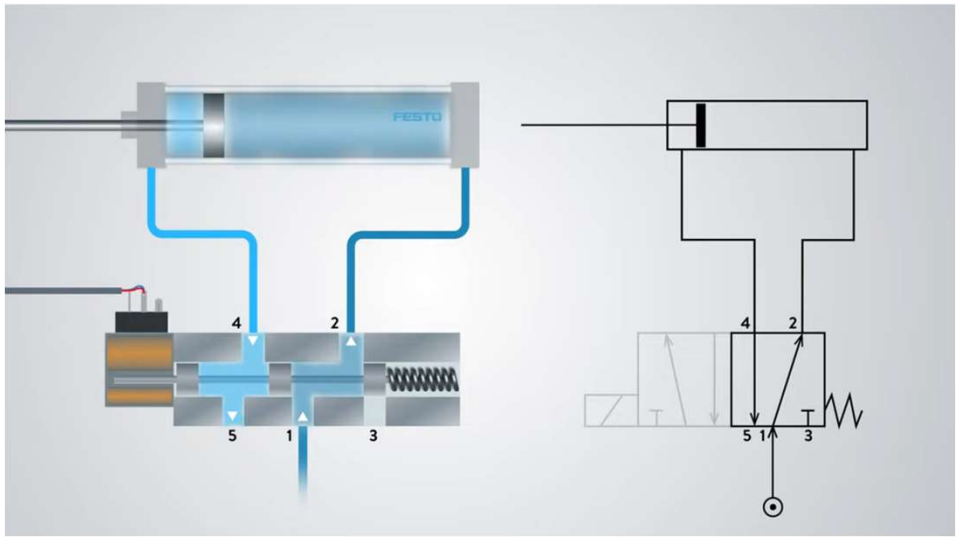


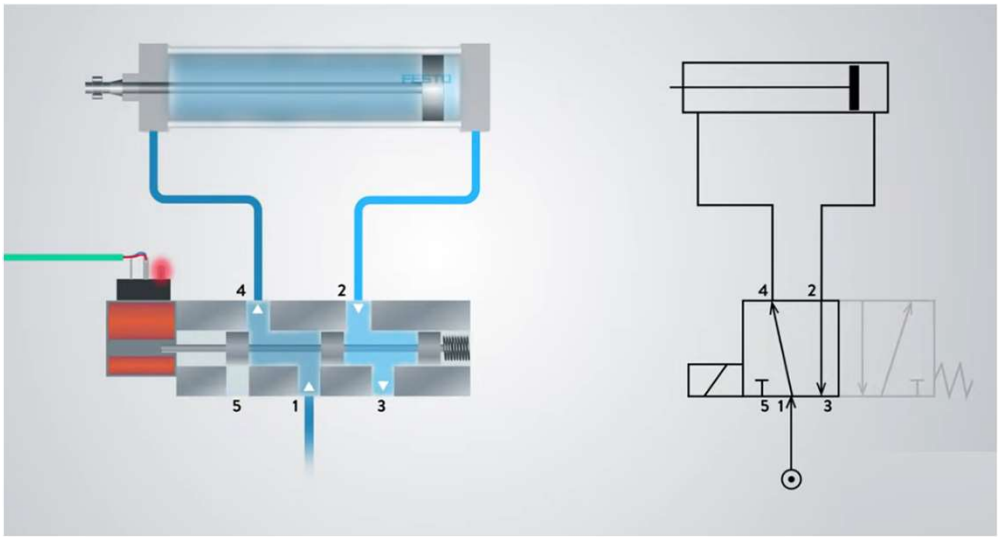
Pneumatic Actuators





Extension and Retraction





Key Advantages & Limitations

Advantages

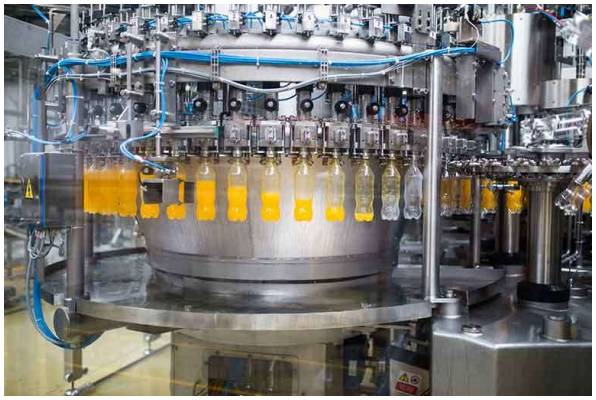
- **Simplicity & Cost :** Air is freely available, and components are relatively inexpensive and simple to maintain.
- **Cleanliness & Safety:**
  - Exhaust air can be released directly into the atmosphere , ideal for food, pharmaceutical, or cleanroom applications.
  - Compressed air does not pose a hazard in terms of fire or explosion.
- **Speed:**
  - Compressed air can flow and exhaust quickly, making it suitable for rapid, repetitive motion (e.g., pick-and-place)
- **Overload Protection:** When overloaded, pneumatic actuators simply stop or stall without damage.
- Cleanliness:

Limitations:

- **Compression:** It is difficult to achieve a uniform and constant piston speed using compressed air.
- **Preparation:** Compressed air must be prepared in advance (e.g., removing dirt and condensed water) to avoid increased wear on components.
- **Force Limit:** Compressed air is only economical up to a certain force requirement.
- **Leakage:** Leakage is an inherent disadvantage tied to high pressure used in pneumatic systems. The chance of leakage increases with the higher pressure.
- **Exhaust Air:** The exhausting air makes a lot of noise.

A few Pneumatics Applications

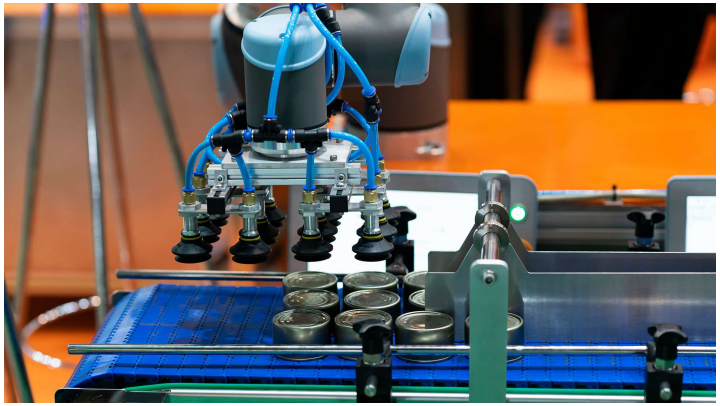
- **Food, beverage or pharmaceutical industries:** Used for tasks like filling, sealing, cutting/dicing, and moving packages.





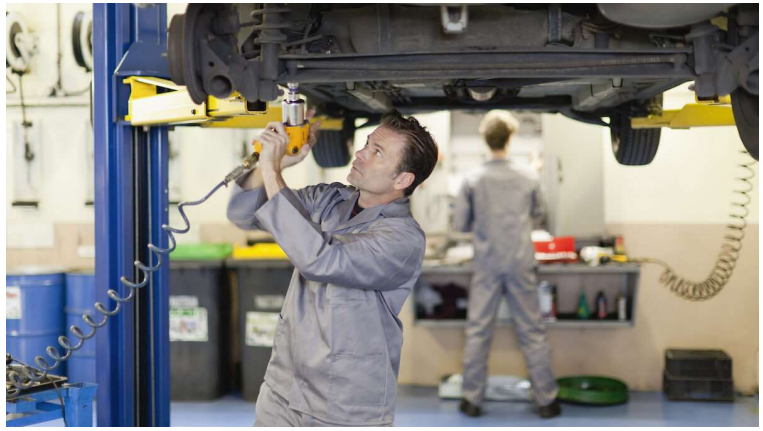
Material Handling & Robotic End-of-Arm Tooling (EOAT)





Workshop / Construction Tools (Hand Tools)

- **Pneumatic jackhammers and nail guns**



Pneumatics Applications

Category	Description	Typical Examples
High-Speed Repetitive Operations	For rapid, continuous actuation where speed and repeatability matter.	Pick-and-place robots, stamping, packaging, ejectors
Clean and Contamination-Free Operations	For hygienic, oil-free, or dust-free tasks where contamination is unacceptable.	Food packaging, pharmaceutical filling, electronics assembly
Moderate-Force / Light-to-Medium Load Applications	For handling and motion where only moderate output force is needed.	Grippers, conveyors, robotic arms, clamps
Safe and Hazardous Environment Operations	For locations where electrical systems are unsafe or explosion-prone.	Chemical plants, mining equipment, flammable processing zones
Portable and Tool-Based Operations	For handheld or mobile equipment using compressed air as a clean power source.	Air drills, grinders, impact wrenches, paint sprayers
Vibration, Lifting, and Transport Systems	For moving, lifting, or vibrating materials safely and smoothly.	Air conveyors, air cushions, pneumatic lifts, air springs

Pneumatics Vs. Hydraulics

Feature	Pneumatic	Hydraulic
Working medium	Air (compressible)	Oil (incompressible)
Pressure range	Low ***(15 – 145 psi)***	High ***(435 – 4350 psi)***
Force capability	Low *(Up to 10 kN (≈ 1 ton) for medium cylinders)*	High *(Up to 500–1000 kN (50–100 tons) or more)*
Precision	**Moderate** (affected by air compressibility)	**High** (incompressible fluid enables fine control)
Speed	**0.1 – 2 m/s** (fast response, quick actuation)	**0.01 – 0.3 m/s** (slower but smoother, more controllable)
Cleanliness	Cleaner, simpler	Requires maintenance, leak control
Component strength	Lightweight, simpler construction	**Heavier-duty**, robust components for high loads